

# **CariSurg MedTech Pathways Research Project**

## **Preliminary Proposal**

**Project Title:** AI-Assisted Triage Validation in a Caribbean Emergency Department: A 12-Week Pilot at Mercer General Hospital

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### **Statement of the Problem**

Triage nurses at Mercer General Hospital contend with high inter-rater variability in a 450-bed Caribbean emergency department that manages seasonal patient surges and significant resource constraints. Existing AI triage models — including TriageIntelli, which achieves 80% accuracy on Korean Triage and Acuity Scale (KTAS) data (Araouchi & Adda, 2024) — have not been validated on Caribbean patient cohorts, and the equity impacts of AI-assisted triage in resource-limited, small-island settings remain unstudied (Da'Costa et al., 2025; Hinson et al., 2025). This proposal describes a 12-week pilot to retrain a stacking ensemble model on de-identified Mercer ED data and evaluate whether AI-assisted triage reduces inter-rater disagreement by at least 15%.

### **Previous Work**

A growing body of literature demonstrates the potential of artificial intelligence (AI) and machine learning (ML) to improve emergency department triage accuracy, reduce mis-triage rates, and support more equitable clinical decision-making. The five peer-reviewed studies summarised below establish the evidence base for this proposal and collectively surface the gaps that a Caribbean-context pilot at Mercer General Hospital would address.

#### **Paper 1 — Tyler et al. (2024)**

Tyler and colleagues conducted a scoping review of peer-reviewed primary studies indexed in EMBASE, Ovid MEDLINE, and Web of Science (2013–2023), examining the effect of AI and ML on triage accuracy, early diagnosis, and clinician workload across US hospital emergency departments. Findings demonstrated measurable improvements in triage accuracy and earlier diagnosis; however, overconfident AI outputs were identified as a patient-safety concern, and the evidence base was found to be heterogeneous. The authors acknowledge that restricting the search to US-based journals limits global generalisability, and note the absence of standardised outcome metrics across included studies. This limitation is directly relevant to Mercer: without Caribbean-context data, the transferability of US-derived AI triage evidence to this setting cannot be assumed.

### **Paper 2 — Da'Costa et al. (2025)**

Da'Costa and colleagues synthesised peer-reviewed articles published between 2015 and 2024 (PubMed, Scopus, IEEE Xplore, Google Scholar) on the benefits, implementation challenges, and future directions of AI-driven ED triage, with particular attention to algorithmic bias and equity. AI systems were found capable of improving patient prioritisation, reducing wait times, and optimising resource allocation; nevertheless, data quality deficiencies, algorithmic bias, limited clinician trust, and ethical concerns were identified as significant barriers to widespread adoption. The authors explicitly note that most reviewed studies originate from high-income settings, leaving applicability to resource-constrained and Caribbean contexts largely unaddressed. This observation directly motivates the equity dimension of the proposed Mercer pilot.

### **Paper 3 — Araouchi & Adda (2024)**

Araouchi and Adda developed and evaluated TriageIntelli, an AI-assisted multimodal triage system for patient prioritisation using the Korean Triage and Acuity Scale (KTAS). Seven ML models — including SVM, Random Forest, XGBoost, and a stacking ensemble — were trained on a single KTAS-labelled Kaggle dataset, with SMOTE oversampling applied to address class imbalance; the stacking ensemble achieved 80.05% accuracy with an AUC of approximately 0.93. The authors identify reliance on a single, culturally specific dataset as the primary limitation, explicitly noting unproven generalisability to settings with different demographics and seasonal disease patterns. The stacking ensemble architecture from this study is the direct technical foundation for the proposed Mercer pilot, which would retrain and validate the same model architecture on local Caribbean data.

### **Paper 4 — Abdalhalim et al. (2025)**

Abdalhalim and colleagues conducted a PRISMA 2020-compliant systematic review of AI-based ED triage systems across four major databases (2020–2025), with six studies meeting inclusion criteria from 119 records; risk of bias was assessed using an adapted QUADAS-2 tool. Voice-AI documentation systems achieved 19% faster throughput compared to manual methods, whilst ML algorithms reduced mis-triage rates by 0.3 to 8.9%, demonstrating tangible clinical impact. The authors flag the predominance of single-centre studies as the primary limitation and explicitly call for multi-centre validation, standardised patient-centred outcome reporting, and equity-focused evidence — none of which has been produced in a Caribbean ED context. The proposed pilot would partially address this call by generating the first such evidence from Mercer.

## **Paper 5 — Hinson et al. (2025)**

Hinson and colleagues performed a secondary analysis of a multisite quality improvement initiative at Johns Hopkins and Yale, evaluating the effect of AI-assisted Emergency Severity Index (ESI) triage on equity outcomes across racial, ethnic, and language-minority subgroups. AI-assisted triage produced measurable improvements in triage equity compared to conventional practice, providing the strongest available evidence that AI decision support can mitigate clinician-level bias in acute care settings. The study is limited by small subgroup sample sizes, the absence of long-term follow-up, and the restriction of all implementation sites to large, well-resourced US academic EDs. The proposed Mercer pilot would test whether these equity benefits extend to a resource-constrained Caribbean context for the first time.

## **Gap Analysis**

### **Gap 1 — Absence of Validation on Caribbean or Small-Island ED Populations**

Araouchi and Adda (2024) trained TriageIntelli exclusively on a Korean KTAS dataset and explicitly identify the absence of validation on demographically or epidemiologically distinct populations as the study's primary limitation. Their model has not been tested on any population outside the original Korean dataset, leaving its performance in Caribbean clinical environments entirely unknown. Mercer's ED serves a population with distinct disease epidemiology — including dengue fever, hurricane-related trauma surges, and locally prevalent non-communicable diseases — not represented in Korean or US training data; deploying a KTAS-trained model without local retraining therefore risks systematic mis-triage. Retraining and validating the stacking ensemble on de-identified Mercer ED records within 12 weeks directly and feasibly addresses the limitation named by the original authors, using an architecture already demonstrated to achieve over 80% accuracy in a comparable supervised classification task.

### **Gap 2 — No Equity Evidence for AI-Assisted Triage in Resource-Constrained Settings**

Hinson et al. (2025) demonstrated that AI-assisted triage improves equity across minority subgroups, but exclusively in large, well-resourced US academic emergency departments; the authors acknowledge that the findings have not been replicated in smaller or lower-resource contexts. Da'Costa et al. (2025) corroborate this gap, noting that most AI triage research draws on high-income-country data and that the applicability of findings to resource-constrained settings — including Caribbean EDs — remains unaddressed. Caribbean emergency departments face staffing constraints, infrastructure limitations, and data-capture variability that may generate bias patterns distinct from those observed in Western academic settings, meaning equity outcomes cannot simply be extrapolated. A pilot that stratifies AI-assisted triage outputs by available demographic variables at Mercer would generate the first equity-focused evidence from a Caribbean ED within the 12-week timeframe, using data already collected as part of routine clinical practice.

## Proposed Solution

The pilot addresses Gap 1 by retraining a validated ML architecture on local Caribbean data, and Gap 2 by measuring equity outcomes in a resource-constrained setting for the first time. It is structured across three sequential phases over 12 weeks.

1. Data extraction and preparation (Weeks 1–3): De-identified triage records will be extracted from Mercer's ED information system. Variables will include presenting complaint, vital signs, assigned triage acuity, disposition, and available demographic fields. Data will be cleaned, class-balanced using SMOTE where necessary, and partitioned into training and held-out test sets.
2. Model retraining and validation (Weeks 4–9): The stacking ensemble architecture from TriageIntelli (Araouchi & Adda, 2024) will be retrained on Mercer data. Performance will be evaluated on accuracy, AUC, sensitivity, and specificity. Inter-rater variability between AI-assisted and nurse-assigned triage will be quantified using Cohen's kappa, directly addressing Gap 1.
3. Equity analysis and reporting (Weeks 10–12): AI triage outputs will be stratified by available demographic variables to assess differential performance across patient subgroups, directly addressing Gap 2 as identified by Hinson et al. (2025) and Da'Costa et al. (2025). Findings will be compiled into a final memo published to the carisurg-portfolio GitHub repository as a PDF.

The primary success criterion is a reduction in inter-rater triage disagreement of at least 15% relative to the pre-implementation baseline. Secondary outcomes include a model AUC of 0.80 or above on the held-out Mercer test set and documented equity metrics stratified by available subgroup variables.

## References

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